

A PROJECT REPORT ON

**Healthcare Application Using Indian Sign
Language (ISL)**

SUBMITTED TO THE SAVITRIBAI PHULE UNIVERSITY, PUNE
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ABSTRACT

Communication is an essential part of human interaction, but individuals with disabilities, particularly those who use Indian Sign Language (ISL), often face challenges due to a lack of platforms that can meet their needs. ISL is more difficult to interpret than American Sign Language (ASL), which primarily uses one hand, because it requires two-handed gestures. Furthermore, the lack of standardized datasets for ISL makes it more challenging to create tools to assist these individuals.

In order to solve these problems, our project, "Healthcare Application Using Indian Sign Language," intends to develop a platform that enables individuals with disabilities to communicate effectively in medical environments.

Our project aims to improve healthcare accessibility, bridge the communication gap, and enhance understanding of the needs of ISL users by promoting inclusivity and awareness. This program is a critical step in fostering an inclusive atmosphere that will make routine healthcare encounters more approachable and encouraging for everyone.

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CHAPTER 1

INTRODUCTION

1.1 Overview

"Healthcare Application Using Indian Sign Language (ISL)" is a project that attempts to improve communication between healthcare professionals and ISL-using patients with speech and hearing impairments. The project addresses the intricacy of ISL, which is more difficult to comprehend than American Sign Language (ASL) because to its two-handed motions and absence of standard datasets. It recognises the dearth of platforms designed specifically for ISL users.

By enabling real-time ISL communication in medical settings, this program will help people with disabilities communicate with medical experts in an efficient manner.

Enhancing healthcare accessibility, encouraging diversity, and fostering a deeper comprehension of the requirements of ISL users are the main goals. The project makes a significant step in creating a healthcare system that is more welcoming and helpful for everybody by doing this.

1.2 Motivation

There is a growing demand in India for a platform that makes it easier for people with disabilities to communicate, particularly in healthcare settings. Lack of standardised Indian datasets for Indian Sign Language (ISL) is a significant obstacle that hinders the development of new technologies in this area. ISL is distinct and requires specific methods for successful interpretation since it involves two-handed gestures as opposed to American Sign Language (ASL), which typically uses one hand.

1.3 Problem Definition and Objectives

People with disabilities in India have trouble communicating because there isn't a standardised, easily accessible platform for Indian Sign Language (ISL), especially in crucial industries like healthcare. The two-handed movements utilised in ISL and the scarcity of large datasets make technical implementation challenging. Additionally, non-ISL users do not have the tools to learn the language, which leads to social exclusion and limited access to essential services.

Objectives:

- To develop a dual-mode communication system using ISL gestures and text/audio input.
- To enable real-time translation of ISL gestures into text and speech.
- To encourage non-sign language users to learn and become more aware of ISL.
- To use inclusive technology to improve healthcare accessibility.

Limitations

- Continuous hand movements can be challenging to accurately comprehend, especially when more complicated expressions in Indian Sign Language are involved.
- Lack of standardised ISL datasets: The lack of standardised datasets specifically designed for Indian Sign Language limits the model's performance and accuracy.
- Other Colour Recognition Tools: To assist those with disabilities in recognising colours, certain tools are needed, such as haptic devices or braille slates, which are not part of the existing platform.

1.4 Methodologies of Problem solving

In order to overcome communication barriers, the research employs Indian Sign Language (ISL) using a systematic strategy that combines deep learning, computer vision, and user engagement. Data acquisition, the initial phase in the process, entails using a webcam to capture video frames. Real-time hand and facial landmark extraction is done with Mediapipe, and the keypoints are stored for later processing. The feature vectors created by compressing these focal points are subsequently utilised as inputs for model training.

During the model training phase, the collected data is divided into training and testing sets. A deep learning model based on the MobileNetV2 architecture is used for gesture recognition. Despite being lightweight, this model performs well on photo categorisation tasks. It features layers for feature extraction, regularisation, and multi-class classification. The model is trained

and validated using the prepared dataset, and its performance is evaluated by evaluating how effectively it can identify different indicators. Once trained, the model is saved and used in real-time applications.

During the execution phase, the trained model is integrated with a webcam interface to enable real-time gesture recognition. Camera frames are preprocessed and then supplied to the model for prediction. When a confidence score and the expected sign are displayed on the screen, the method becomes interactive and simple to use.

Another element of the system is a Text to Hand Sign Conversion module, which makes it easier to turn text-based pharmaceutical instructions into hand sign movies. By mixing relevant predetermined hand motion GIFs according to user input, including time and dosage, a thorough instructional film is produced. This tool makes medical advice more accessible to ISL users, especially in cases when conventional instructions, like taking medication, are involved.

All things considered, this technology ensures a dependable and efficient solution for ISL-based communication in healthcare by fusing interactive visual outputs with gesture detection.

CHAPTER 2

LITERATURE SURVEY

Literature Survey

The authors of the work by Sharma et al. focus on recognising English alphabets from sign language using machine learning techniques. It adds a great deal to the field, but because it only covers the English alphabet, its scope is limited. This restriction limits its applicability to other sign languages, such as Indian Sign Language (ISL), which has a distinct collection of signs and grammatical structures.

The Kanchana Devi et al. study's authors look into gesture recognition for those with physical impairments. Their study tackles the challenge of managing a large vocabulary, which frequently presents a barrier to sign language recognition. Additionally, as the report glosses over the details of sign language recognition for people with disabilities, there is room for additional research in this area.

Satti et al. present an audio-to-text-to-sign language converter. Despite its potential, this strategy's failure to satisfy ISL standards is a significant drawback. Any conversion system for sign language must accurately capture the unique vocabulary, grammar, and cultural background of ISL. If ISL standards were not followed, the system's effectiveness and utility for the Indian deaf community would be limited.

According to the literature review, machine learning techniques for sign language recognition are becoming more and more popular. There is clearly a need for research that addresses the challenges of distinguishing different sign languages, including ISL, and considers the particular needs of those with disabilities.

CHAPTER 3
SOFTWARE REQUIREMENTS
SPECIFICATION

3.1 Assumptions and Dependencies

In this section, we study the Assumptions and Dependencies of our project, user classes and characteristics along with assumptions and dependencies of our project.

This project assumes that users are familiar with basic computer functions and have access to a webcam-enabled device with internet connectivity. Furthermore, the application assumes that users or carers are either learning or already skilled in Indian Sign Language (ISL).

The dependencies are as follows:

- The model works only with a trained CNN based on MobileNetV2.
- TensorFlow, OpenCV, and more Python libraries must be installed.
- The availability and consistency of datasets are essential for efficient training and evaluation.
- Webcam functionality is necessary for real-time gesture recognition.

3.2 Functional Requirements

3.2.1 Gesture Recognition

The system should be able to capture live webcam footage and identify preset ISL motions linked to a number of medical conditions, including cough, fever, and cold.

3.2.2 Text to Hand Sign Conversion

The system should convert text input (such "Morning Full") into a sequence of gesture videos for medication regimens and dosages.

3.2.3 Real-Time Feedback

The system would provide real-time visual feedback (predicted sign and confidence level) to ensure user clarity and model performance.

3.2.4 Multi-Device Compatibility

The system is designed to function on a range of platforms, including PCs and laptops running Windows, Linux, or macOS. Due to its minimal processing requirements and webcam access, it can be used on standard hardware without the need for a GPU.

3.3 External Interface Requirements

3.3.1 User Interfaces

- A simple GUI displaying gesture predictions in real time
- Input text box or buttons for converting text to gesture
- GUI to display the Text to Hand sign conversion

3.3.2 Hardware Interfaces

- Webcam (integrated or external) for gesture capture
- Display for showing the processed output

3.3.3 Software Interfaces

1. Operating System: Windows / Linux / macOS
2. Development Tools: Jupyter Notebook / VS Code
3. Dataset pertinent to medical images
4. Programming Language: Python
5. Libraries: OpenCV, TensorFlow, NumPy, Mediapipe, MobileNetV3
6. Model File: `Final_handSignModel.h5`

3.3.4 Communication Interfaces

- Local execution with webcam input.
- Optional: Future versions may integrate APIs for voice-to-text conversion.

3.4 Nonfunctional Requirements

3.4.1 Performance Requirements

- The system should respond to gesture input within 1–2 seconds.
- Lightweight enough to run on standard machines with 8GB RAM.

3.4.2 Safety Requirements

- Should not cause system crashes or data corruption.
- Should handle camera access exceptions gracefully.

3.4.3 Security Requirements

- Local system execution ensures data privacy.
- No user data is stored or transmitted externally.

3.4.4 Software Quality Attributes

- **Usability:** Easy-to-use GUI
- **Reliability:** Consistent performance across sessions
- **Maintainability:** Modular code for future extension

3.5 System Requirements

3.5.1 Database Requirements

- Currently, the system is file-based.
- JSON format is used for storing landmark data during training.
- Future versions may use MongoDB or SQLite for gesture metadata.

3.5.2 Software Requirements (Platform Choice)

- Python 3.9+
- Jupyter Notebook / IDE
- Required Libraries: OpenCV, TensorFlow, NumPy, Mediapipe

3.5.3 Hardware Requirements

- Webcam
- Minimum 8GB RAM
- Intel i5 / AMD Ryzen 5 or above

3.6 Analysis Models: SDLC Model to be applied

The project follows the **Agile SDLC model**, allowing iterative development and continuous feedback, given the need for continuous testing and improvement of the gesture recognition models.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

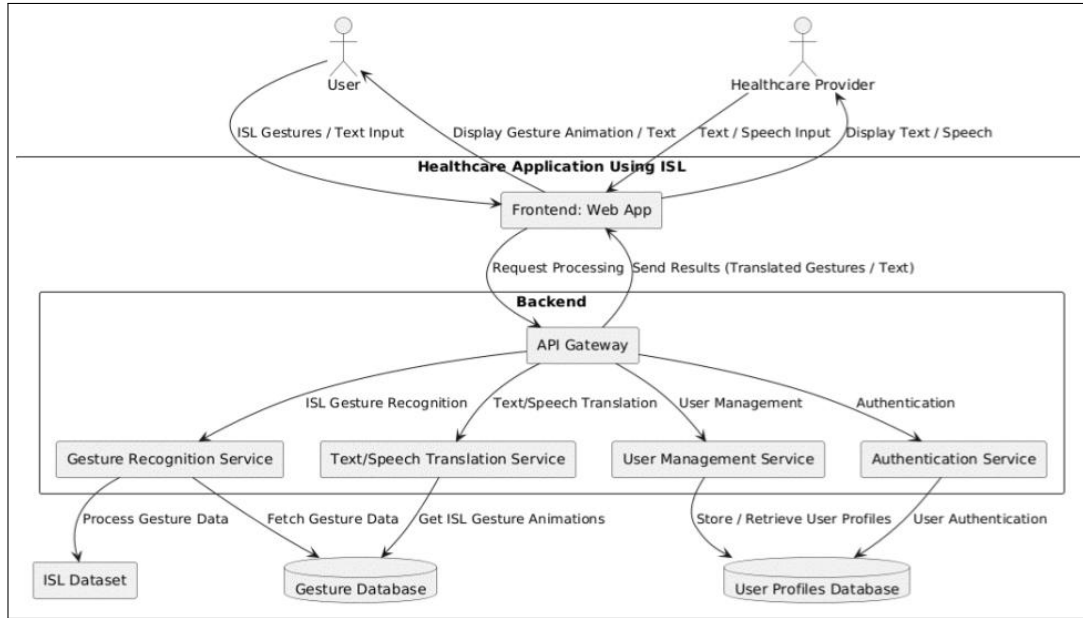


Figure 4.1: System Architecture

4.2 Mathematical Model

This section provides a formal mathematical representation of the bidirectional ISL communication system using **Finite Automata (FA)** models for each translation direction.

1. Finite Automaton for Sign Language to Text (FA)

Definition: $FA = (Q, \Sigma, q, F)$

Where:

- $Q = \{q, q_1, q_2, q_3, q_4, q_f\}$ — Set of states
- $\Sigma = \{\text{video frame, features, gesture, text}\}$ — Input alphabet
- $\delta : Q \times \Sigma \rightarrow Q$ — Transition function
- q — Initial state

- $F = \{q_f\}$ — Final/accepting state

State Descriptions:

- q : System idle, waits for gesture input
- q : Captures video frame
- q : Extracts hand landmarks (MediaPipe/OpenCV)
- q : Classifies the gesture using MobileNet model
- q : Converts recognized gesture to corresponding text/audio
- q_f : Displays or outputs the result

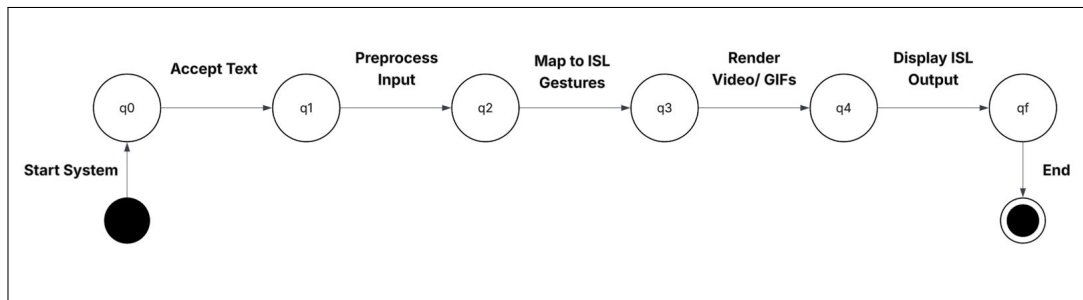


Figure 4.2: Finite Automata Model for Bidirectional ISL Communication

2. Finite Automaton for Text to Sign Language (FA)

Definition: $FA = (Q, \Sigma, p, F)$

Where:

- $Q = \{p, p_1, p_2, p_3, p_4, p_f\}$ — Set of states
- $\Sigma = \{\text{text/audio, word, ISL-sign, 3D-render}\}$ — Input alphabet
- $\delta : Q \times \Sigma \rightarrow Q$ — Transition function
- p — Initial state
- $F = \{p_f\}$ — Final/accepting state

State Descriptions:

- p: System waits for input from user/doctor
- p: Processes input (text/audio)
- p: Splits sentence into words
- p: Maps each word to corresponding ISL sign
- p: Renders gesture using 3D Blender model
- pf: Displays gesture animation

Language Accepted:

- FA accepts ISL gesture sequences and outputs corresponding text/audio.
- FA accepts textual/audio input and outputs animated ISL gestures.

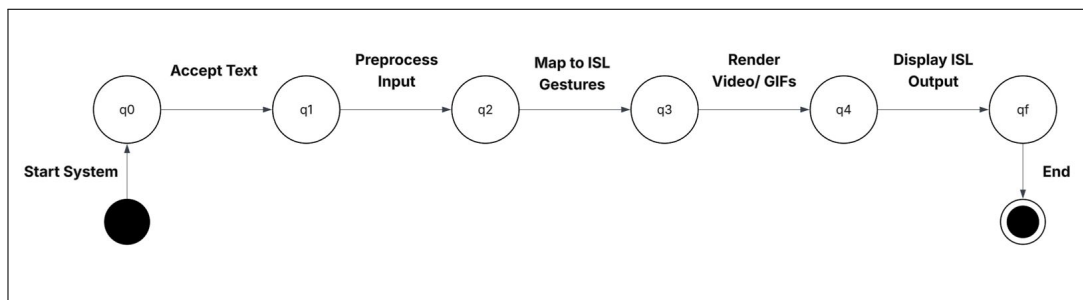


Figure 4.3: Finite Automata Model for Bidirectional ISL Communication

4.3 Data Flow Diagrams

Every major program function is explained, together with either the data flow (structured analysis) or the class hierarchy (analysis class diagram with class description for object-oriented systems).

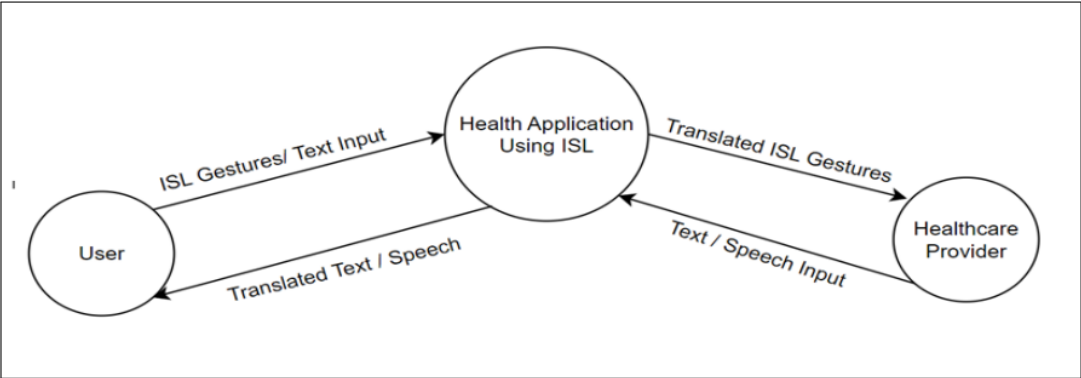


Figure 4.4: Data Flow Diagram - Level 0

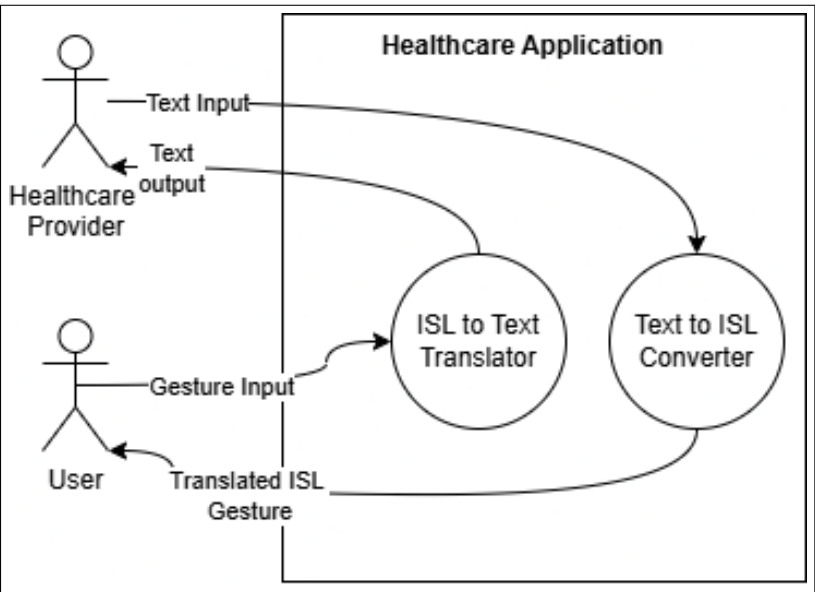


Figure 4.5: Data Flow Diagram - Level 1

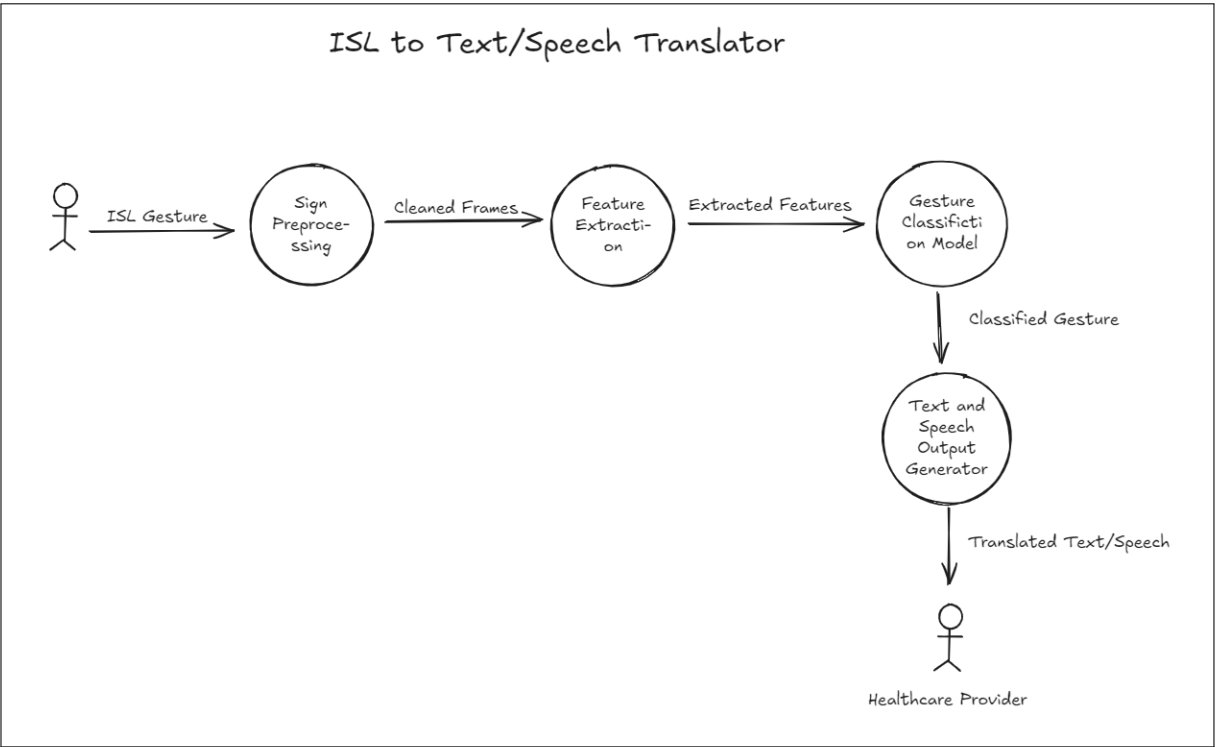


Figure 4.6: Data Flow Diagram - Level 2

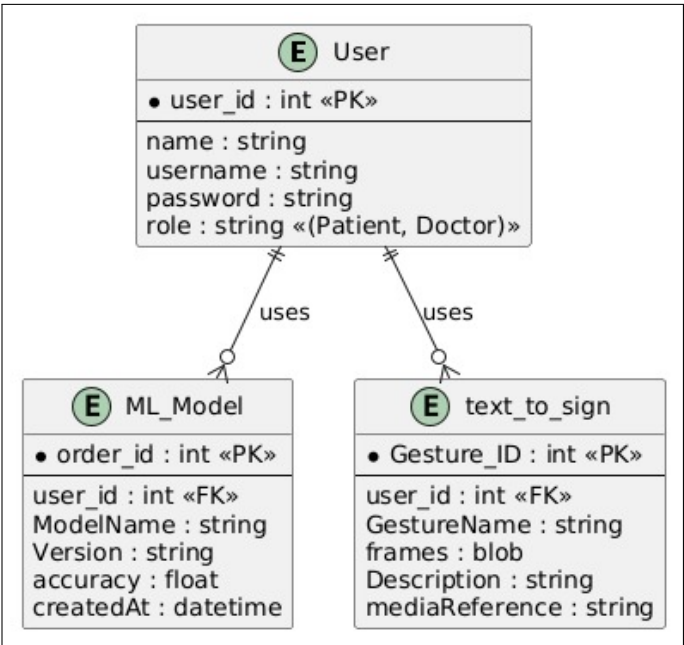


Figure 4.7: Class Diagram

4.4 Class Diagram

Here, we have shown the class diagram

4.5 UML Diagrams

The following figure represents the use-case diagram of the system.

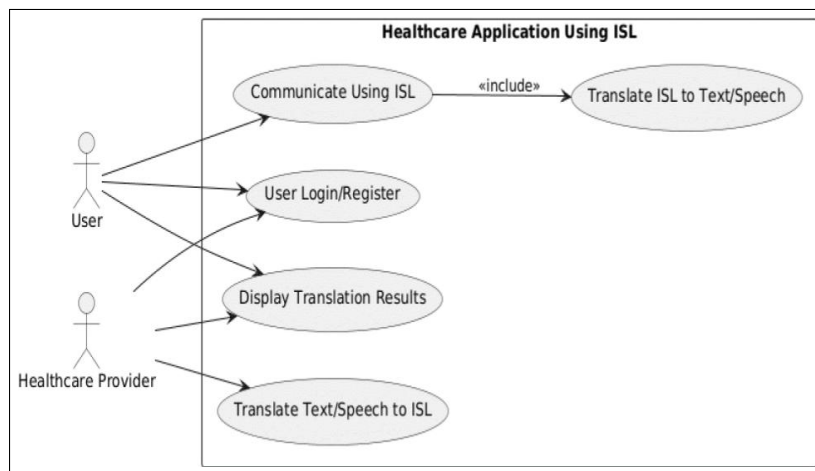


Figure 4.8: Use-Case Diagram

4.6 Sequence Diagram

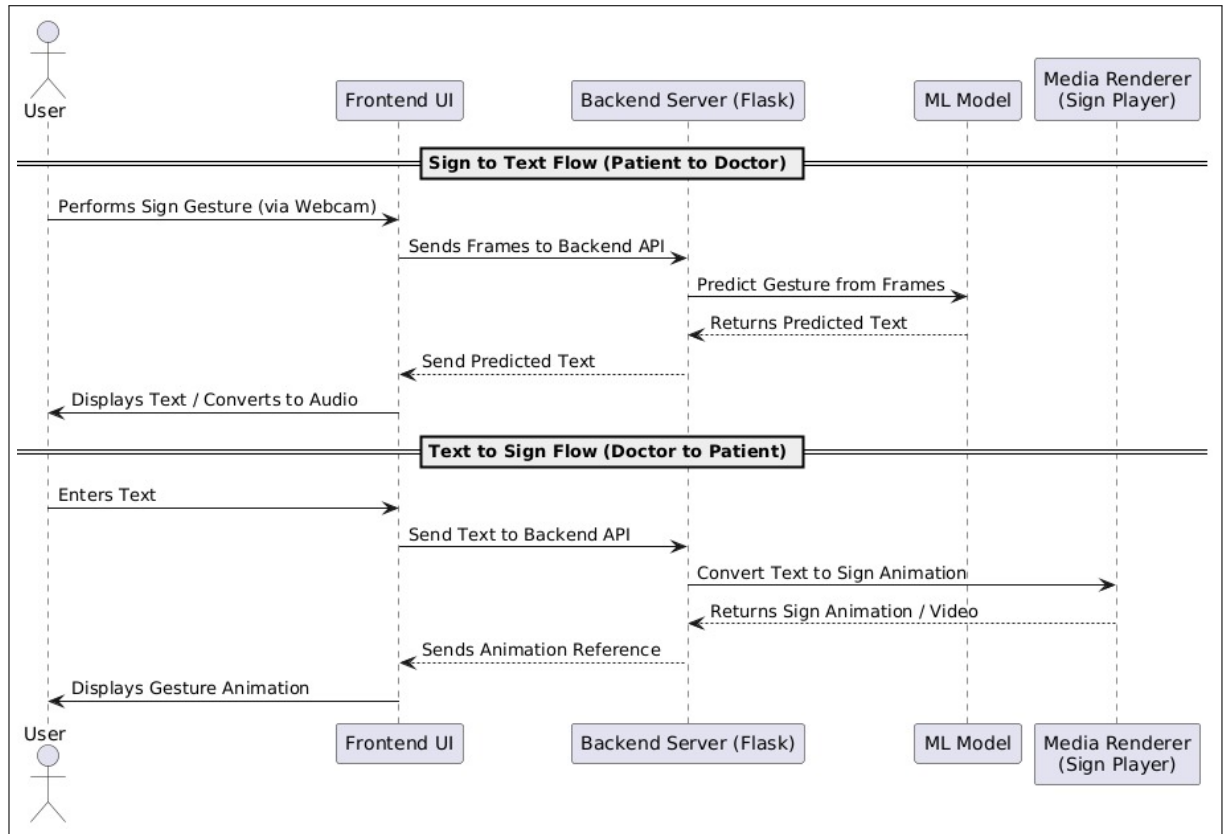


Figure 4.9: Sequence Diagram

CHAPTER 5

PROJECT PLAN

5.1 Project Estimate

5.1.1 Reconciled Estimates

- The amount of time needed to construct and train the gesture recognition model.
- The creation of text-to-gesture and ISL-to-text capabilities.
- Including an ISL learning module to help users.
- Analysis of user input, documentation, testing, and debugging.

5.1.2 Project Resources

- **Human Resources:** 4 team members managing development, UI, testing, and documentation.
- **Software Tools:** Python, TensorFlow, OpenCV, Jupyter Notebook.
- **Hardware:** Laptops with webcams, 8GB+ RAM.
- **Reference Material:** Research papers and open-source ISL datasets.

5.2 Risk Management

- A comprehensive approach to identify, assess, and mitigate risks to ensure a project is developed and completed effectively.

5.2.1 Risk Identification

- There aren't many ISL gesture datasets available.
- Recognition errors brought on by gesture overlap.
- Delays or errors in processing videos in real time.
- Issues with hardware or operating system compatibility between computers.

5.2.2 Risk Analysis

- Dataset insufficiency – **High Probability, High Impact**
- Model underperformance – **Medium Probability, High Impact**
- Real-time delay – **Medium Probability, Medium Impact**
- Hardware issues – **Low Probability, Medium Impact**

5.2.3 Overview of Risk Mitigation, Monitoring, Management

- **Mitigation:** Using open-source datasets and manual augmentation.
- **Monitoring:** Weekly testing sessions and performance monitoring.
- **Management:** Contingency planning and substitute models for crucial attributes

5.3 Project Schedule

5.3.1 Project Task Set

- **Phase 1: Research and Requirement Gathering** (Month 1)
Completion of requirements and scope, analysis of competition, and examination of literature.
- **Phase 2: Dataset Acquisition and Preprocessing** (Month 2)
To include the collection, labelling, cleaning, and standardisation of ISL datasets.
- **Phase 3: Model Development** (Month 3 and 4)
The gesture recognition model is created and trained using MobileNetV2, and the prediction accuracy is initially evaluated.
- **Phase 4: Integration and Testing** (Month 5)
Unit tests, text-to-sign, gesture detection, and ISL learning modules are connected, and the GUI is put in place.
- **Phase 5: Final Testing, Deployment, and Feedback** (Month 6)
Complete system testing, deploy, fix defects, and gather user and guidance feedback. Responses and Implementation

5.3.2 Task Network

- Research and Requirement Gathering → Dataset Preprocessing
- Dataset Preprocessing → Model Development
- Model Development → Integration
- Integration → Testing
- Testing → Deployment and Feedback

5.3.3 Timeline Chart

- Month 1: Research and Requirement Gathering
- Month 2: Dataset Acquisition and Preprocessing
- Month 3–4: Model Development
- Month 5: Integration and Testing

- Month 6: Final Testing, Deployment, and Feedback

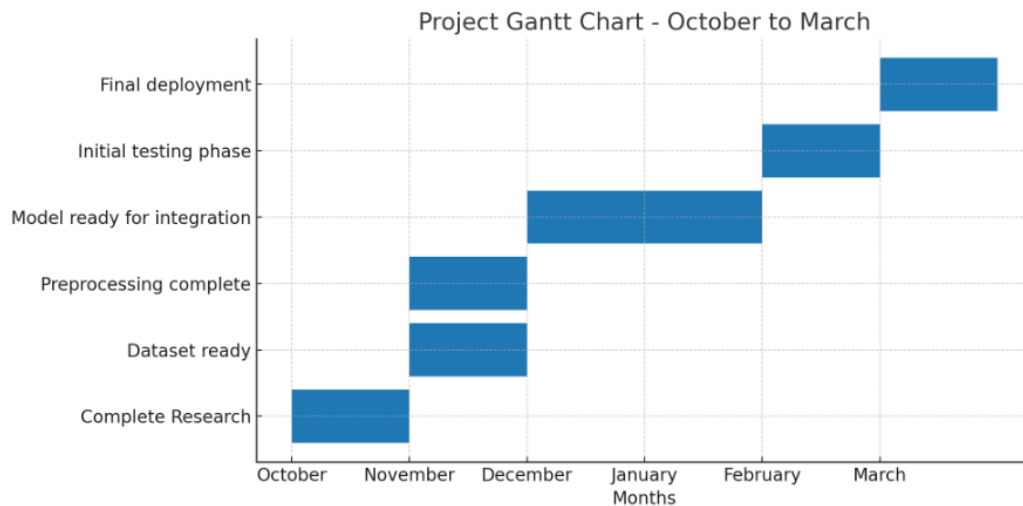


Figure 5.1: Gantt Chart for Project Milestones and Timeline (2024–2025)

5.4 Team Organization

5.4.1 Team structure

- **Four Developers:** Responsible for implementation of gesture detection, Text to Hand sign conversion and UI testing.
- **Project Guide:** Manages communication with faculty, monitors progress, and ensures milestones are met.

5.4.2 Management reporting and communication

- Weekly internal team meetings for status updates and task division.
- Monthly reports shared with project guides (Prof. Preeti Jain and Prof. Aarti Deshpande).
- Milestone-based documentation submissions after each project phase.
- Final presentation and project review with feedback from all stakeholders prior to deployment.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

Several interrelated modules are used in the Healthcare Application Using Indian Sign Language (ISL) implementation:

- **Gesture Recognition Module:** Uses a webcam to record live video input and recognises pre-established ISL signs associated with medical problems.
- **Text-to-ISL Converter:** This tool uses a mapping database to translate user-provided health-related text input into associated gesture movies.
- **Real-Time Feedback Interface:** Provides visual representation, accuracy/confidence score, and gesture prediction in real-time feedback.

6.2 Tools and Technologies Used

The following tools and technologies were employed during the implementation phase:

- **Programming Language:** Python 3.9+
- **Libraries:** OpenCV, TensorFlow, NumPy, Mediapipe
- **Deep Learning Framework:** TensorFlow/Keras
- **Model Architecture:** MobileNetV2-based CNN for gesture classification
- **IDE:** Jupyter Notebook, Visual Studio Code
- **Hardware:** Standard laptop with webcam, 8GB RAM, and Intel i5 processor or comparable
- **Dataset:** Custom ISL dataset captured for common health phrases

6.3 Algorithm Details

6.3.1 Algorithm 1: Hand Landmark Detection Using Mediapipe

Input: Real-time video stream from webcam

Output: Set of hand landmarks (keypoints)

1. Capture a frame from the video stream.
2. Apply Mediapipe's hand detection model.
3. Extract 21 hand landmark coordinates (x, y, z).
4. Normalize the coordinates for input to the CNN classifier.

Usage: This algorithm feeds processed hand landmarks to the gesture classification model.

6.3.2 Algorithm 2: ISL Gesture Classification Using CNN

Input: Preprocessed hand landmark data

Output: Predicted ISL gesture (e.g., 'Cough', 'Fever')

1. Load the trained MobileNetV2-based model.
2. Pass the landmark vector through the CNN layers.
3. Use softmax activation to output class probabilities.
4. Select the gesture with the highest confidence score.

Usage: The predicted gesture is displayed in the GUI and used for text-to-sign conversion.

6.3.3 Algorithm 3: Text to Gesture Video Conversion

Input: User-input health-related text

Output: Video sequence of corresponding ISL gestures

1. Tokenize the input sentence into words or phrases.
2. Match each token with a predefined gesture video in the database.
3. Concatenate and display the gesture sequence for user comprehension.

Usage: Enables caretakers or users to visually understand health-related text content in ISL.

CHAPTER 7

SOFTWARE TESTING

Type of Testing

- **Unit Testing:** Each core module, such as Sign-to-Text Conversion, Text-to-Sign (using 3D avatar), and Camera Frame Processing, was tested independently to ensure isolated functionality.
- **Integration Testing:** All modules including webcam input, model prediction, audio generation, and sign rendering were integrated and tested together to confirm seamless data flow and coordination.
- **System Testing:** The entire application was deployed and tested on target devices to verify overall functionality in real-time healthcare-related scenarios.
- **User Acceptance Testing (UAT):** The system was evaluated by end users including differently-abled individuals, faculty, and students. Feedback focused on gesture recognition accuracy, 3D sign rendering clarity, and ease of communication with doctors.
- **Functional Testing:** Confirmed that all intended features—such as capturing signs via webcam, converting signs to meaningful text/audio, converting doctor's input to ISL gestures, and ISL learning modules performed as described in the Software Requirements Specification (SRS).
- **Performance Testing:** Verified that the system processed camera frames and responded with predicted output (text/audio/sign) within 1–2 seconds under standard network and hardware conditions, ensuring real-time usability.

7.1 Test cases & Test Results

Test ID	Test Case Description	Expected Output	Actual Output	Status
TC 01	Webcam gesture capture	Live gesture stream visible in window	Gesture feed captured correctly	Pass
TC 02	Recognize ISL gesture for “Cough”	“Cough” predicted with confidence score	“Cough” (95%) displayed	Pass
TC 03	Convert text “Morning Full” to ISL gestures	Sequence of gestures played for “Morning Full”	Correct gesture video sequence played	Pass
TC 04	No gesture in frame	No prediction or “No gesture detected”	Message “No gesture detected” displayed	Pass
TC 05	System load with continuous usage (30 mins)	No crash; consistent gesture prediction	System stable; consistent performance	Pass

Table 7.1: Test Cases and Results for ISL Healthcare Application

CHAPTER 8

RESULTS

8.1 Outcomes

1. successfully created a dual-mode communication platform that uses Indian Sign Language (ISL) to enable communication between healthcare providers and people with disabilities.
2. Using camera input, real-time ISL gesture detection was implemented, increasing accessibility and communication by giving users precise feedback and visual confirmation.
3. To promote diversity and awareness, a text-to-sign conversion capability was integrated to help carers, students, and other users comprehend and learn ISL.

8.2 Screen Shots



Figure 8.1: Gesture Recognition

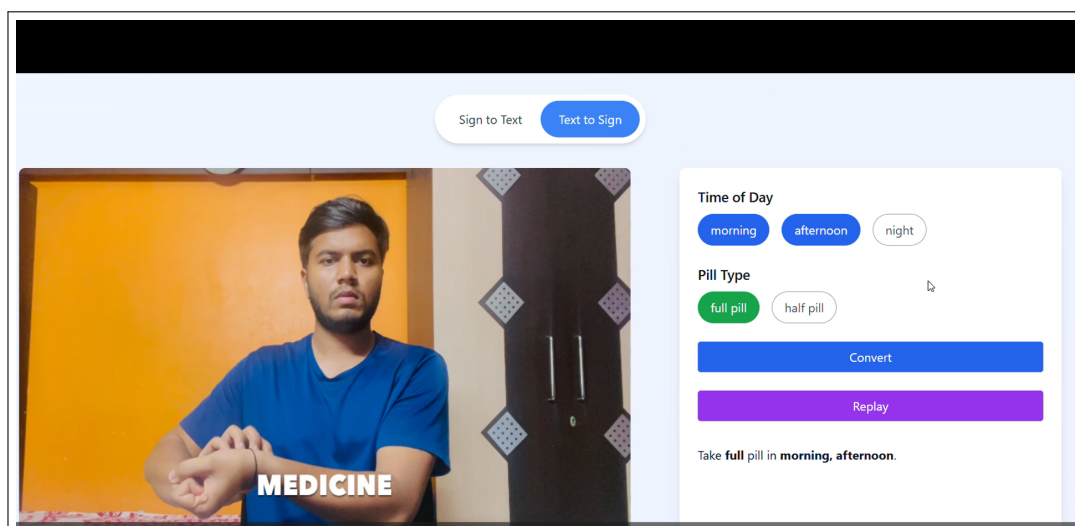


Figure 8.2: Text to Hand Sign Conversion

CHAPTER 9

CONCLUSIONS

9.1 Conclusions

1. Using Indian Sign Language (ISL) and an intuitive application, the project effectively closes the communication gap between healthcare providers and people with disabilities.
2. Comprehensive usability for patients and carers is ensured by the combination of text-to-sign translation, real-time ISL gesture recognition, and an ISL learning module.
3. The system's use of computer vision and machine learning allowed it to predict gestures accurately and with little latency, which makes it appropriate for real-time healthcare situations.

9.2 Future Work

1. Adding more sophisticated medical and conversational phrases to the ISL gesture lexicon will increase expressiveness.
2. Creation of a mobile application version for easier smartphone access and wider distribution.
3. Integration of cloud storage and synchronisation capabilities for analytics and gesture logs in institutional configurations.
4. For improved regional flexibility throughout India, multilingual support and regional ISL variations should be included.

9.3 Applications

- Hospitals and Clinics – for real-time communication with differently-abled patients.
- Rehabilitation Centers – to support sign language learning and therapy.
- Educational Institutions – for teaching ISL to caregivers and students.
- NGOs and Support Groups – to promote inclusive healthcare communication.
- Home Care Settings – where caretakers can use the app to understand patient needs more efficiently.

CHAPTER 10

APPENDIX

10.1 Appendix A

This section evaluates the feasibility of the proposed project titled “**Health-care Application Using Indian Sign Language (ISL)**” by analyzing the system from the perspectives of logical satisfiability, computational complexity (P, NP, NP-Complete, NP-Hard), and modern algebraic models relevant to the architecture and functionality of the application.

1. Satisfiability Analysis

Satisfiability analysis involves verifying whether a set of conditions or constraints within a system can be logically fulfilled. In the context of this project, each ISL gesture must accurately correspond to a specific output in the form of text or audio, and vice versa. The success of the system depends on whether these gesture-to-output mappings can be consistently satisfied given a set of input variables such as hand position, finger orientation, motion, and timing.

The problem can be modeled as a Boolean Satisfiability (SAT) problem, where the goal is to determine if there exists an assignment of truth values to variables that satisfies a given Boolean formula. For example, the system must determine whether a combination of hand coordinates matches a specific ISL gesture. If such mappings exist for all supported gestures, the problem is satisfiable.

In essence:

- A **satisfiable system** indicates that the application can consistently translate gestures into meaningful outputs.
- An **unsatisfiable system** suggests that the conditions or model design may need to be altered to ensure functional reliability.

Satisfiability analysis thus supports the feasibility of achieving gesture recognition and translation under real-world constraints like lighting, background noise, and hand movement precision.

2. Computational Complexity Analysis (P, NP, NP-Complete, NP-Hard)

Understanding the complexity of gesture recognition and translation tasks is essential for determining the scalability and efficiency of the application.

Task	Complexity Class	Justification
Preprocessing video frames, extracting hand landmarks	P (Polynomial Time)	These operations using MediaPipe or OpenCV are computationally efficient and feasible for real-time applications.
Gesture classification involving ambiguous signs	NP (Non-deterministic Polynomial Time)	Solutions can be verified quickly, but computing the correct classification may be difficult due to overlapping or uncertain gesture inputs.
ISL-to-text translation under noisy or dynamic input	NP-Complete / NP-Hard	Real-time recognition of complex, overlapping gestures and multi-modal inputs introduces significant computational challenges similar to optimization problems.

Table 10.1: Computational Complexity Classification of ISL Healthcare Application Tasks

Despite potential NP-completeness in edge cases, practical implementations using machine learning models and feature extraction reduce the real-time computational burden, making the application usable and efficient.

4. Conclusion

The proposed project is practically feasible within its defined scope. Logical satisfiability ensures gesture recognition can be reliably implemented, while complexity theory informs performance considerations in real-time use cases. Algebraic models enhance system structure, enabling consistent and logical gesture-to-text/audio mappings. With appropriate optimizations and learning models, the application can serve as a robust solution for Indian Sign Language-based healthcare communication.

10.2 Appendix B: Details of Paper Publication

Paper Title: *Healthcare Application Using Indian Sign Language*

1. Name of the Conference/Journal

Conference/Journal Name: *International Journal of Engineering, Management & Humanities [IJEMH]*

Publication Link: https://ijemh.com/issue_dcp/Healthcare%20Application%20Using%20Indian%20Sign%20Language.pdf

2. Reviewers' Comments

Reviewer 1: "This paper addresses a socially significant problem by proposing a healthcare communication aid for differently-abled individuals using Indian Sign Language. The integration of computer vision and deep learning makes the concept both innovative and practical. Some diagrams or illustrations of the system pipeline could enhance clarity. "

Reviewer 2: "The paper presents a well-structured overview of technologies like YOLOv5, CNN, and MediaPipe used for ISL recognition. The relevance to the healthcare sector is clearly highlighted. However, a comparison with existing solutions or benchmarks would strengthen the evaluation. "

Reviewer 3: "The focus on bidirectional communication for medical interactions is timely and important. The authors have done a commendable job summarizing the potential of sign language recognition in healthcare. Future work suggestions could benefit from more technical depth."

3. Paper

Citation:

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4. Certificate

Certificate of Publication: https://ijemh.com/issue_certificate/553624.pdf

10.3 Appendix C: Plagiarism Report

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ABSTRACT Communication is an essential part of human interaction, but individuals with disabilities, particularly those who use Indian Sign Language (ISL), often face challenges due to a lack of platforms that can meet their needs. ISL is more difficult to interpret than American Sign Language (ASL), which primarily uses one hand, because it requires two-handed gestures. Furthermore, the lack of standardized datasets for ISL makes it more challenging to create tools to assist these individuals. In order to solve these problems, our project, "Healthcare Application Using Indian Sign Language," intends to develop a platform that enables individuals with disabilities to communicate effectively in medical environments. Our project aims to improve healthcare accessibility, bridge the communication gap, and enhance understanding of the needs of ISL users by promoting inclusivity and awareness. This program is a critical step in fostering an inclusive atmosphere that will make routine healthcare encounters more approachable and encouraging for everyone.

CHAPTER 1 INTRODUCTION

CHAPTER 11

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LIST OF ABBREVIATIONS

Abbreviation	Illustration
ISL	Indian Sign Language
ASL	American Sign Language
CV	Computer Vision
CNN	Convolutional Neural Network

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